

32 2025-04-21 | Week 14 | Lecture 32

Chapter 14.1 through 14.3

We will finish the example of a test of independence from last time.

Example 144 (Test of independence). For example, suppose we have a population of $n = 54$ students, which we divide up according to two factors

- (i) whether they attended class regularly or not
- (ii) whether they pass the class exam or not

Then our contingency table is

	Pass	Fail
Attended	25	6
Skipped	8	15

(23)

This is the *table of observed values*. We usually augment the table with “marginal totals”, like so:

	Pass	Fail	Total
Attended	25	6	31
Skipped	8	15	23
Total	33	21	54

The null hypothesis is

$$H_0 = \text{attendance and passing are independent}$$

and the alternative hypothesis is

$$H_a = \text{there is a relationship between attendance and passing.}$$

Let p denote the probability that a student regularly attends class, and let q denote the probability that a student passes the final exam.

Important: We do not know the true value of p and q . But we can estimate them using the marginal row and column sums from the table:

$$p \approx \frac{31}{54} \quad \text{and} \quad q \approx \frac{33}{54}$$

Question: What are the expected values of the table, assuming the null hypothesis is true?

Under the null hypothesis, we can multiply probabilities, e.g., so that the probability that a student both regularly attends class AND passes the exam is about pq . Therefore, since there are n students,

$$\begin{aligned}
 & \mathbb{E}[\# \text{ of students who attend regularly AND pass the exam}] \\
 &= n\mathbb{P}[\text{a randomly-selected student attends regularly AND pass the exam}] \\
 &= n\mathbb{P}[\text{attends regularly}] \mathbb{P}[\text{passes the exam}] \\
 &= npq
 \end{aligned}$$

Similar calculations gives us the following table of expected counts:

	Pass	Fail
Attended	npq	$np(1-q)$
Skipped	$n(1-p)q$	$n(1-p)(1-q)$

and plugging in the values $n = 54$ and our approximations $p \approx \frac{31}{54}$ and $q \approx \frac{33}{54}$ gives

	Pass	Fail
Attended	18.9	12.1
Skipped	14.1	8.9

(24)

Interpretation: These are the values we would expect if H_0 were true. If the observed values deviate too much from these values, that's evidence against the null hypothesis.

Using the values recorded in Tables 25 and 26, we can compute the chi-squared statistic:

$$\begin{aligned}\chi^2 &= \sum_{\text{all table entries}} \frac{(\text{observed value} - \text{expected value})^2}{\text{expected value}} \\ &= \frac{(25 - 18.9)^2}{18.9} + \frac{(6 - 12.1)^2}{12.1} + \frac{(8 - 14.1)^2}{14.1} + \frac{(15 - 8.9)^2}{8.9} \\ &= 11.9\end{aligned}$$

By Remark 141, we know that χ^2 has a chi-squared distribution. When computing a chi-squared test statistic from a contingency table for a test of independence, the degrees of freedom is given by the equation

$$\text{df} = (\text{number of rows} - 1) \times (\text{number of columns} - 1)$$

which in our case, for a 2×2 table, is just 1.

Therefore, using Definition 137, the p-value is

$$\begin{aligned}p\text{-value} &= \mathbb{P}[\chi^2 \geq 11.9] \\ &= \int_{11.9}^{\infty} \frac{1}{\sqrt{2\pi}} x^{-\frac{1}{2}} e^{-x/2} dx \\ &= 0.0006\end{aligned}$$

In other words, if attending and passing were independent, we would see results this extreme only 0.06% of the time. This is very small, so we reject the null hypothesis. Our conclusion is that attendance and passing the final exam are **not independent**, i.e., that there is a relationship between them.

At this point we are tempted to conclude that regularly attending class increases your chance of passing the final exam. This seems obvious and is certainly consistent with the data. But it's not a conclusion that follows from the test that we did. The test only told us that these two things probably have some relationship; it doesn't give us information about the nature of that relationship.

End of Example 144. $\square$

32.1 Interpreting tests of independence

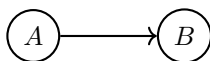
Suppose we want to know if two things A and B are related to each other. We will discuss only the case where A and B are discrete, categorical events. For example

A = [a person vapes]

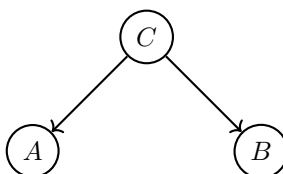
B = [person gets coronary artery disease (CAD)]

There are many possible relationships:

- **Causality:** Vaping causes arterial disease.

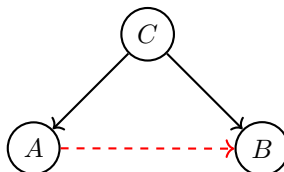


- **Common Response:** Perhaps the effect of vaping on your arteries is perfectly benign (no causal relationship), but CAD and vaping are still positively associated due to the effect of some third variable C that causes both to occur simultaneously:



For example we might have $C = [\text{person smokes cigarettes}]$. Maybe vapers are disproportionately likely to have smoked cigarettes, and perhaps it's this subset of vapers that drives the association between vaping and CAD, as cigarette smoking is well-established to cause CAD. If this is true, then quitting vaping wouldn't decrease your chance of CAD.

- **Confounding:** (A third variable jointly influences or causes both A and B). For example, suppose it were observed that vapers have **lower** rates of coronary artery disease than the general population — that doesn't mean vaping necessarily has a protective effect. For example, maybe the vapers are on average younger than the population (CAD usually develops in older people). In that case, we might observe a spurious relationship between A and B due to the effects of some other third variable, $C = [\text{youth}]$:



It could nonetheless be the case that vaping increases your risk of CAD through some biological mechanism (the red dotted line), and due to the effect of age, vapers might *still* have lower incidence of CAD than the general population.

In order to rule this out, we'd need to *control* for the effect of age, perhaps by conducting a study on people from the same age group.

What we *can* say, however, is that if A and B do not have any relationship (causal or otherwise), then they are probabilistically independent. This is what tests of independence do. They ask the question “**Are A and B independent?**” The answer to such tests is always either “no” or “maybe”. They don't give us much more information than that.